# AI-Powered Job Application Agent

## Final Project Presentation - Automating Resume Tailoring and Application Submission


Team Swift

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Date: December 5, 2025

# Project Overview, Alignment & Core Workflow

## Section 1: Initial Project Idea & Charter Description

Idea: Our core idea is to have an AI platform/tool that Automates resume tailoring based on the Job Description (JD). The platform/tool will also apply for the job.

Project Charter: "This project will deliver a SaaS platform that enables an AI Agent to apply for jobs on behalf of users. The aim is to simplify and accelerate the job-seeking process while enhancing personalization and efficiency."

Alignment: This innovative solution expands our market leadership in HR technology by focusing on application efficiency and personalization.

## Section 2: Core Workflow (Prioritized from Whiteboard)

The process is focused on intelligently altering the Master Resume by focusing only on the Skills and Experience the user has provided.

# Project Charter - Scope & Objectives

**Project Sponsor:** ZipRecruiter

**Project Team:** All team members here

**Expected Start Date:** 2025-10-01

**Expected Completion Date:** 2026-06-30 (9-month duration)

## Key Requirements (Usable Features):

**Secure User Authentication**
Secure User Authentication and Profile Management.

**AI Tailoring**
AI-powered resume and cover letter tailoring.

**Board Integration**
Automated integration with at least 3 job boards (e.g., ZipRecruiter, LinkedIn, Indeed, Workday, Greenhouse, SmartRecruiters, YCombinator, Jobright.ai).

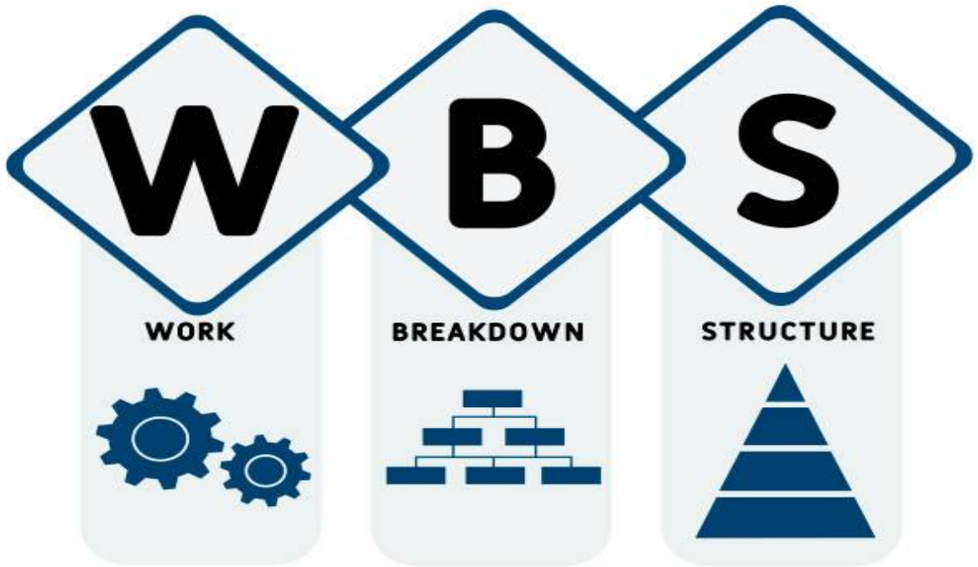
**Smart Matching**
Smart job matching using user preferences.

**Tracking Dashboard**
Application tracking dashboard.

**Compliance & Privacy**
Compliance safeguards for data privacy.

## Project Exit Criteria:

- Working SaaS platform released by June 30, 2026.
- AI Agent successfully applies across at least 3 major job boards.
- Resume/cover letter tailoring validated by pilot users.



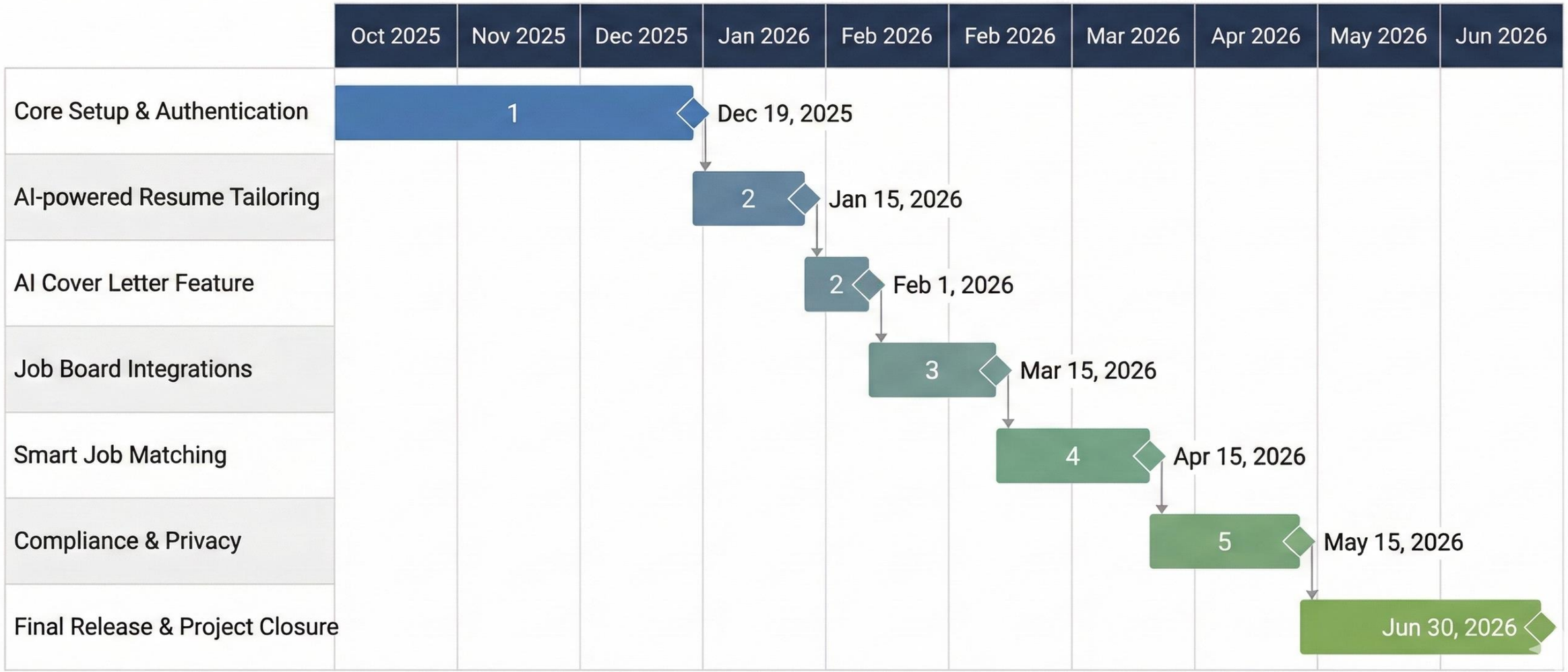

# Work Breakdown Structure (WBS)

The project is organized into 5 functional releases, followed by Testing & Release.

| Unique ID | WBS Level | Breakdown (WBS Item) | Primary Deliverables |
|---|---|---|---|
| 1 | Release 1 | Core Setup – Agent & Authentication | UI, Auth (OAuth2/SSO), MFA, Profile Management |
| 2 | Release 2 | AI Resume & Cover Letter Tailoring | AI Bullet Edits, UI Preview |
| 3 | Release 3 | Job Board Integrations (Pilot Scope) | API/Web Scrapping Research, Job Posting Retrieval, Auto-Submission |
| 4 | Release 4 | Matching & Tracking | Smart Job Matching Algorithm, Tracking Dashboard Design |
| 5 | Release 5 | Privacy & Compliance | Data Privacy Safeguards, Security Testing |
| 6 | Release 6 | Testing & Release | End-to-End Testing, UAT, Final Launch |

# Project Schedule (Gantt Chart Milestones)

**Total Project Duration: 9 Months (Oct 2025 – Jun 2026)**



# Budgetary Estimate - Part A

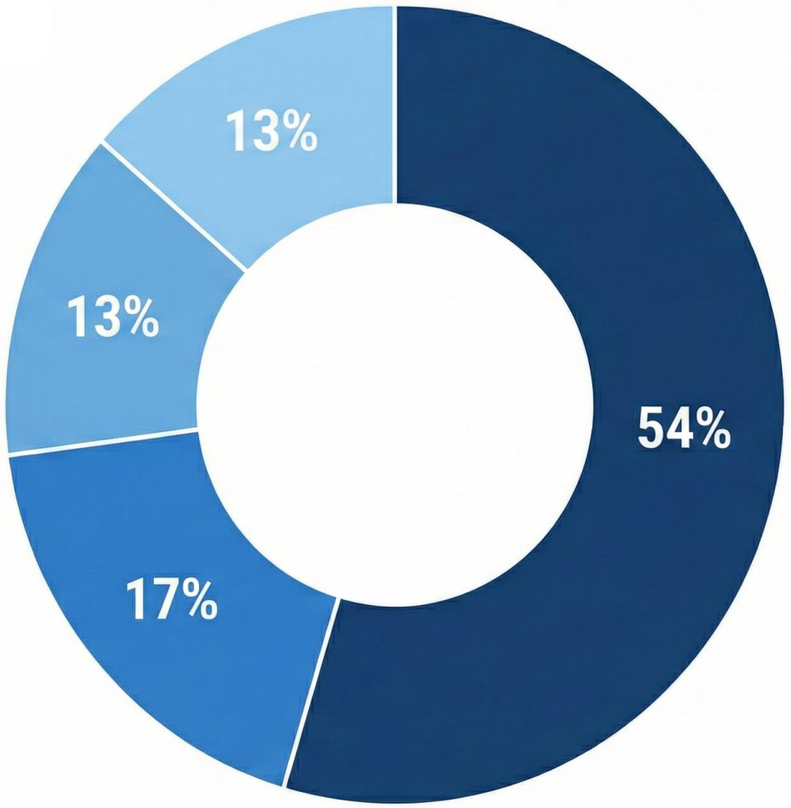
**Total Project Budget (MVP): $478,488**

| Budget Item | Units/Basis | Cost/Unit | Total Costs | Notes |
|---|---|---|---|---|
| Labor | 3,150 Hours | $83 / hr | $261,488 | Blended rate across roles (Dev, PM, QA, Security). |
| Tools & Infrastructure | 1 | $65,000 | $65,000 | Increased for AI API tokens and and cloud hosting. |
| Hardware | 6 Units | $2,500 | $15,000 | High-performance workstations for AI Model dev. |
| Training | 1 | $5,000 | $5,000 | Security & Compliance workshops. |
| Overhead/G&A | 20% of Labor | — | $52,290 | Facility, utilities, and HR costs. |
| **Subtotal** | — | — | **$398,740** | — |
| Contingency Reserves | 10% of Subtotal | — | $39,874 | Held by PM for known risks. |
| Management Reserves | 10% of Subtotal | — | $39,874 | Held by Business for unknown risks. |

# Labor Allocation Breakdown

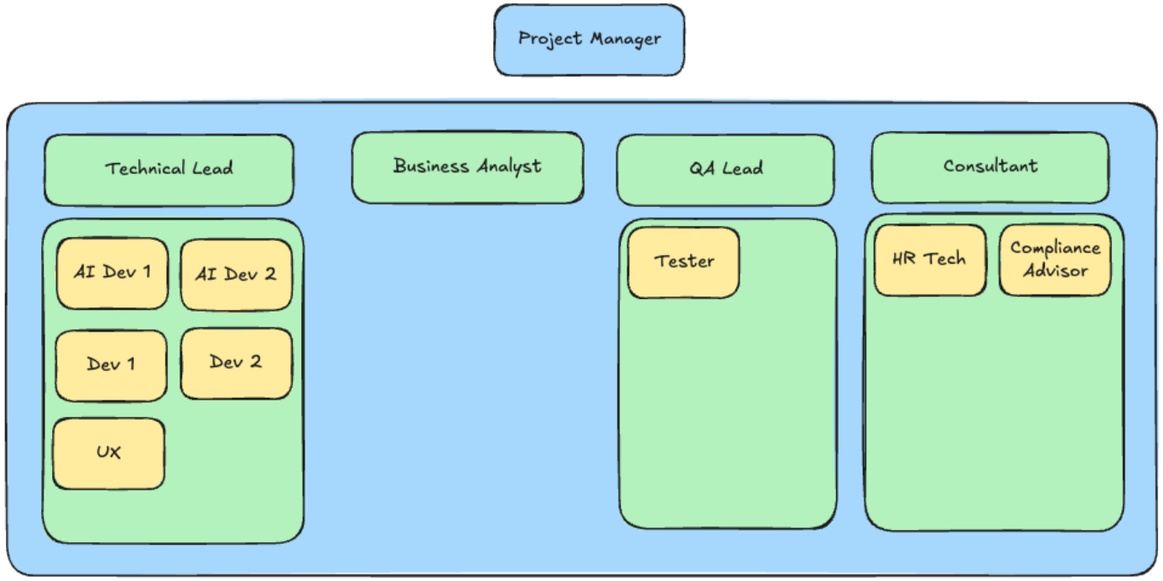
**Below is a detailed breakdown of the labor allocation across different role categories for the project, showcasing both total hours and the percentage of effort contributed by each team.**

**Total Hours: 3,150**
Percentage breakdown across role categories.




- Developers (AI & Full Stack) - 1,690 Hours
- Project Leads & Planning - 530 Hours
- Testers (QA) - 400 Hours
- Designers & Security - 420 Hours

# Team Organization Chart & Resources




## RACI Matrix Highlight (AI Resume Tailoring):

> **Accountable (A):** Project Manager, Technical Lead.**Responsible (R):** AI Developer 1.**Consulted (C):** AI Developer 2, Developer 1, Business Analyst.**Informed (I):** UX Designer, Developer 2.

# Make-Buy Analysis

**Deliverable Focus:** Compliance & Data Privacy Safeguards (Release 5)

**Decision:** MAKE (Internal Development)

**Total Cost to MAKE:** $20,500

**Total Cost to BUY:** $57,500

Create a professional comparison table:

| Cost Item | Cost to MAKE | Cost to BUY |
|---|---|---|
| Initial SME Setup | $14,500 | $39,500 (Consultants) |
| Ongoing Support (Q4-Q2) | $6,000 | $18,000 |

**Rationale:** Making the solution internally builds necessary internal expertise for long-term maintenance and is the significantly most cost-effective option for the project.

# Test Cases and Quality Assurance Findings

**Scenario Focus:** AI Resume & Cover Letter Tailoring (Release 2)

Initial test results indicate the core AI functionality requires immediate attention.

| Unique ID | Test Case | Actual Result Summary | Status | Key Action Required |
|---|---|---|---|---|
| 1 | Verify AI tailoring page is displayed correctly. | Page loaded correctly; UI sections visible. | ✅ Passed | UI Shell is ready. |
| 2 | Verify AI-generated suggestions are relevant to the JD. | AI only generated a section mapping skills ("ADDITIONAL DETAILS"), but did not tailor the core resume content. | ⚠️ Partially Passed | Rework AI Model for deep content personalization as specified in the project idea. |
| 3 | Verify users can review, edit, and save the tailored resume. | Resume saved successfully; but the editor was a static document editor (like MS Word), not the expected interactive editor. | ⚠️ Partially Passed | Rework UI Editor to provide the modern, interactive experience required for a no-code SaaS platform. |

## Key QA Conclusion:

The current AI output fails to deliver on the core requirement to alter the resume. Release 2 requires rework to ensure the AI's depth and the UI's editing experience meet user and project expectations.

# SWOT Analysis

STRENGTH

Sleek and clean UI, very easy to onboard users to use the job agent portal, is a strong selling point.

WEAKNESS

High dependence on the quality and stability of external job board APIs.

OPPORTUNITY

Partnering with Universities and Coding Bootcamps to license the platform for student career centers.

THREAT

Evolving Anti-Bot Protocols: Job boards frequently update security (CAPTCHAs, Rate Limits) to block generic bots. Mitigation: Our focus on "Human-in-the-Loop" review ensures our traffic mimics human behavior.

# Stakeholder Communications Plan

Communication is focused on mitigating the highest risks and managing stakeholder expectations.

## Engagement Strategy (Stakeholder Assessment Matrix):

| Stakeholder | Current State | Desired State | Key Focus |
|---|---|---|---|
| Project Sponsor | Supportive | Leading | Consistent reporting of budget and schedule adherence. |
| Job Boards | Resistant | Neutral | Formal communications focusing on ethical, controlled data access. |
| Consultant | Unaware | Neutral | Focused workshops and information sharing on compliance strategy. |

## Key Communication Events:

| Situation | Channel | Frequency | Owner | Internal Audience |
|---|---|---|---|---|
| Weekly Status | Team Stand-up + Email Report | Weekly (Every Thursday) | Deputy PM | Project Team (Dev, QA, BA, AI) |
| Monthly Review | Virtual Review + Dashboard | Monthly (1st Monday) | Project Manager | Technical Lead, BA, Sponsor Finance |
| Product Launch | Company Townhall + Email | One-time (2026-06-30) | Project Manager | All Internal Staff |

# Summary & Next Steps

**Project Status:** Planning complete. Budget ($478k) and Schedule (9 Months) are set.

**Critical Finding:** Initial testing shows the UI Editor and AI Tailoring depth need rework (Release 2) to meet user expectations.

**Ask:** We request approval of the budget and the plan to prioritize the Release 2 rework immediately to ensure the product's core value proposition is met.

**Q&A:** Open floor for questions.